

the packet loss occurs. If ping does not return a reply, it means that the network communication is not established. On success, the ping command exits with code 0. Otherwise, it will exit with code 1 or 2. This can be useful when using the ping utility in a shell script.

traceroute - Print the route packets trace to a network host

traceroute command in Linux prints the route that a packet takes to reach the host. This command is useful when you want to know about the route and about all the hops that a packet takes. Syntax:

```
traceroute [options] host_Address [pathlength]
```

Options: -4 Option: Use ip version 4 i.e. use IPv4. Syntax:

```
$ traceroute -4 10 google.com
```

-6 Option: Use ip version 6 i.e. use IPv6. Syntax:

```
$ traceroute -6 10 google.com
```

-F Option: Do not fragment packet. Syntax:

```
$ traceroute -F google.com
```

-f first_ttl Option: Start from the first_ttl hop (instead from 1). Syntax:

```
$ traceroute -f 10 google.com
```

-g gate Option: Route the packet through the gate. Syntax:

```
$ traceroute -g 192.168.43.45 google.com
```

-n Option: Do not resolve IP addresses to their domain names. Syntax:

```
$ traceroute -n google.com
```

-p port Option: Set the destination port to use. Default is 33434. Syntax:

```
$ traceroute -p 20292 google.com
```

-q nqueries Option: Set the number of probes per each hop. Default is 3.

Syntax: \$traceroute -q 1 google.com

netstat - Print network connections, routing tables, interface statistics, masquerade connections, and multicast memberships

netstat is a command-line network utility that displays network connections for Transmission Control Protocol, routing tables, and several network interface and network protocol statistics.

Examples of some netstat commands with practical examples are as follows:

```
# netstat -a | more : To show both listening and non-listening sockets.
```

```
# netstat -at : To list all TCP ports
```

```
# netstat -au : To list all UDP ports
```

```
# netstat -l : To list only the listening ports
```

```
# netstat -lt : To list only the listening TCP ports
```